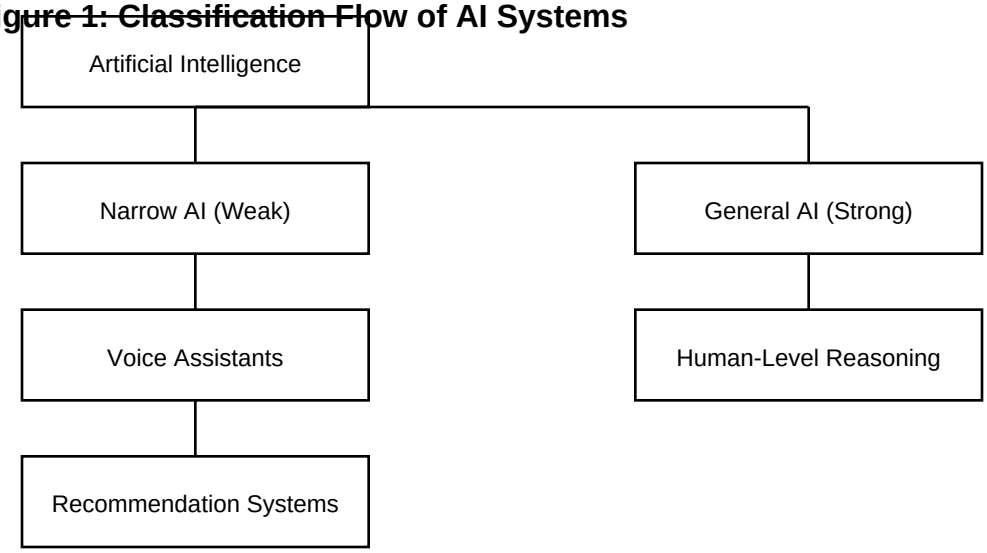
Unit 1: Introduction to Artificial Intelligence

Artificial Intelligence, commonly abbreviated as AI, refers to the simulation of human intelligence in machines that are programmed to think, learn, and make decisions. The field draws on computer science, mathematics, linguistics, and cognitive psychology to build systems capable of performing tasks that would normally require human intelligence, such as visual perception, speech recognition, decision-making, and language translation. Modern AI systems are typically built using machine learning techniques, where a model learns patterns directly from large amounts of data rather than following manually programmed rules. This chapter introduces the core classifications of AI systems, from narrow, task-specific systems to the broader, more general forms of intelligence that remain largely theoretical today.

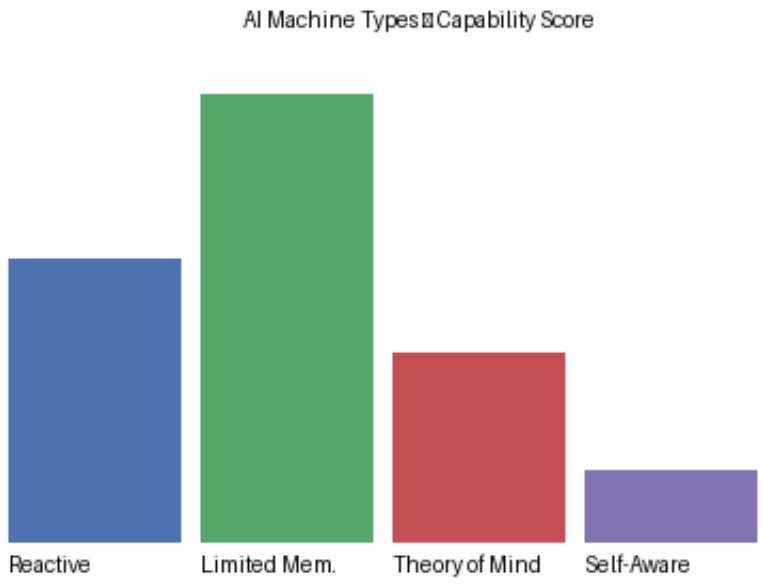
Figure 1: Classification Flow of AI Systems



As shown above, Narrow AI systems power most real-world applications in use today, while General AI remains an active area of research with no fully realized implementation yet.

Figure 2: Capability Comparison Chart

The chart below compares capability scores across machine types.



This concludes Unit 1's introductory material.